

LangChain

LangChain is a framework used to build applications using Large Language Models. It connects LLMs with tools, memory, APIs, and external data so developers can create intelligent systems like chatbots and AI agents.

LLMs (Large Language Models)

LLMs are AI systems trained on massive amounts of text data to understand and generate human-like language. They are used for tasks like answering questions, writing text, translation, and powering chatbots such as ChatGPT.

RAG (Retrieval-Augmented Generation)

RAG is a method that combines information retrieval with language models. It first retrieves relevant data from external sources like documents or databases, then uses an LLM to generate more accurate and context-aware answers.

FAISS (Facebook AI Similarity Search)

FAISS is a library designed for fast similarity search over vector data. It helps in finding related items like text or documents by comparing embeddings, and is widely used in recommendation systems and RAG-based applications.

Vector

A vector is a list of numbers that represents data such as words, sentences, or images in mathematical form. In AI, vectors help convert real-world information into a format that machines can understand and process.

Vector Database

A Vector Database is a system that stores and manages vector embeddings. It enables fast similarity search so AI applications can quickly find related information. Examples include Pinecone, Weaviate, and ChromaDB.

Generative AI (GenAI)

Generative AI refers to artificial intelligence systems that can create new content such as text, images, music, or code. Unlike traditional AI, it focuses on generating new outputs based on learned patterns from data.

GANs (Generative Adversarial Networks)

GANs are deep learning models made of two networks. The generator creates fake data, while the discriminator tries to detect whether it is real or fake. This competition improves the quality of generated outputs like images and videos.